

Eigenvalues and inverse problems

The sharp spectral-matching constant for normal matrices

Difficulty: extreme **Importance:** interesting to the community

Status: open; historical source, bounded later-status search.

Problem statement

For normal matrices $A, B \in \mathbb{C}^{n \times n}$, list their eigenvalues $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_n$ with algebraic multiplicity. Define the bottleneck matching distance

$$d_\infty(A, B) = \min_{\pi \in S_n} \max_{1 \leq j \leq n} |\alpha_j - \beta_{\pi(j)}|,$$

where S_n is the permutation group. Determine the exact value of

$$C_{\text{normal}} = \sup_{n \geq 2} \sup_{\substack{A, B \in \mathbb{C}^{n \times n} \text{ normal} \\ A \neq B}} \frac{d_\infty(A, B)}{\|A - B\|_2}.$$

Here normal means $AA^* = A^*A$, and $\|\cdot\|_2$ is the spectral norm. The target is a single sharp constant valid in every dimension. Equivalently, find the smallest C for which $d_\infty(A, B) \leq C\|A - B\|_2$ for all such pairs, with matching sharpness evidence. The conjectured value $C = 1$ has already been refuted; it is not the question being posed.

Why it matters

This is a sharp conditioning question for the complete spectrum of a normal matrix. Matching counts repeated eigenvalues, so a Hausdorff-distance estimate that ignores multiplicity does not answer it.

References

R. Bhatia, *Perturbation Bounds for Matrix Eigenvalues*, SIAM, 2007, Chapter IX, pp. 154–155; the question is restated as Problem 27 in X. Zhan, *Open problems in matrix theory*, §21, pp. 12–13. P. Šemrl, *Geometrical techniques in linear algebra*, ICTP lecture notes, 2009, p. 10, Theorem 5 and the following open question. J. A. Holbrook, *Spectral variation of normal matrices*, LAA 174 (1992), 131–144.

Status check

The cited sources distinguish the unresolved sharp constant from Holbrook's counterexample to one. They give $1 < C_{\text{normal}} < 3$. A. Parusiński and A. Rainer, *Eigenvalue stability of Hermitian and normal matrices*, v1 (2026), Proposition 3.4, still states the classical universal bound with $1 < C < 3$; it does not determine the sharp constant. Searches for *sharp spectral variation constant normal matrices*, *spectral variation best possible constant*, and 2025/2026 follow-ups found no exact value. The explicit open-question sources are historical; the recent bound is context, not an explicit reaffirmation of openness. This is a bounded literature check.